

Standard Operating Procedure ***Tube Notcher***

Grizzly G0767 - 4" x 79" Abrasive Tube Notcher

The Grizzly G0767 is a specialized abrasive machining tool designed for high-speed notching of pipe and tubing. It utilizes a 2 HP motor to drive a 4”x 79" abrasive belt at approximately **5900 FPM** (Feet Per Minute). This machine is intended for precision structural fitment and is not a substitute for a general-purpose sander.

Potential Hazards

- **Mechanical**: High-speed belt travel (5900 FPM), pinch points between the abrasive belt and grinding roller, and rotating pulleys.
- **Operational**: High-velocity spark streams, extreme friction heat in the workpiece, and metallic dust.
- **Fire**: Heavy spark generation can ignite nearby combustibles or wood dust.
- **Projectile**: Workpieces can be "kicked" or ejected if the vise is not properly tensioned.

Personal Protective Equipment (PPE)

PPE Item	Requirement
Tie Back Hair & Secure Loose Items	Tie back long hair and secure loose clothing that could get caught in rotating parts. Remove rings, watches, and other jewelry.
Eye/Face Protection	Wear ANSI Z87.1-compliant safety goggles. A face shield may be necessary to protect from high-velocity dust and debris.
No Gloves Advised	STRICTLY PROHIBITED: Gloves can get caught in the abrasive surface or between the table and disc. Use push sticks or boards and follow the "4 inch" rule
Hearing Protection	Mandatory: The combination of a 3450 RPM motor and abrasive contact on hollow tubing creates decibel levels requiring protection.
Protective Clothing	Close-fitting protective clothing, such as a shop apron, is encouraged.
Footwear	Appropriate enclosed footwear must be worn at all times.



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Before Use

- **Belt Inspection**: Ensure belt is free of tears or frays. Verify the belt is tracking centrally on the rollers.
- **Roller Selection**: Ensure the grinding roller matches the Outside Diameter (OD) of the tubing being notched.
- **Vise Setup**: Verify the quick-release vise is adjusted to the correct angle 0 to 30 and the locking levers are fully engaged.
- **Guard Placement**: Ensure the transparent eye shield and the side-access belt guards are closed and secured.

During Use

- **Focus**: Concentrate on the task at hand; avoid distractions. Most injuries occur when the operator is not paying full attention.
- **Startup**: Stand to the side and allow the belt to reach full speed before engaging the stock.
- **Clamping**: **All workpieces must be secured in the vise.** Never attempt to hand-hold tubing during a notch.
- **Feeding**: Use the handwheel to feed the stock into the belt slowly. Do not force the material; let the abrasive remove metal at a steady rate to prevent belt glazing or motor stall.
- **The "4-Inch" Rule**: Keep hands at least 4 inches away from the abrasive contact point.
- **Thermal Management**: Metal tubing becomes extremely hot during abrasive notching. Handle with caution once removed from the vise.
- **The Stop Rule**: Wait for the belt to come to a **complete stop** before attempting to clear debris or adjust the angle.
- **Stalling**: If the machine binds or the belt stops, turn the power off immediately and wait for a complete stop before investigating.

After Use

- **Power Down**: Shut off the motor and ensure the machine is ready for the next technician.



- **Coast-Down Safety**: Wait for the belt to stop on its own—**do not** attempt to use your hands or a scrap piece to stop it.
- **Debris Removal**: Use a brush to collect metal fines from the vise and base once cool. **Avoid compressed air** to prevent driving grit into the machine's bearings.
- **Hygiene**: Wash hands and exposed skin thoroughly to remove metallic dust and abrasive grit.

Emergency Equipment and Procedures

- **Emergency Stop**: Know the location of the large paddle-style E-stop on the control box.
- **Fire Safety**: In the event of a spark fire, use a Class D or ABC fire extinguisher as appropriate for the material.
- **Reporting**: Report all incidents, belt failures, or near-misses to the Laboratory Project Supervisor immediately.